Tropical proof systems

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March 27, 2024

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Good old LP systems

Axioms $x_i \geqslant 0, \;\; 1 - x_i \geqslant 0$.

Cutting Planes (CP) [W.Cook, Coullard, Turán]

$$\frac{f_1 \geqslant 0 \qquad f_2 \geqslant 0}{\alpha_1 f_1 + \alpha_2 f_2 \geqslant 0}\ (\alpha_1, \alpha_2 \geqslant 0), \qquad \frac{\sum_i ca_i x_i - d \geqslant 0}{\sum_i a_i x_i - \lceil d/c \rceil \geqslant 0}\ (c, a_i, d \in \mathbb{Z}).$$

Resolution over Cutting Planes (Res(CP)) [Krajíček]

$$\frac{f_1 \geqslant 0 \ \vee\ \Gamma \qquad f_2 \geqslant 0 \ \vee\ \Gamma}{\alpha_1 f_1 + \alpha_2 f_2 \geqslant 0 \ \vee\ \Gamma}\ (\alpha_1, \alpha_2 \geqslant 0) \qquad \text{(+RES)}$$

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$$\frac{\text{—}}{f - 1 \geqslant 0 \ \vee\ -f \geqslant 0}. \qquad \text{(NEG)}$$

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$$\frac{\text{—}}{x - 1 \geqslant 0 \ \vee\ -x \geqslant 0}\ (x \text{ is a variable}). \qquad \text{(NEG-x)}$$

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Good old LP systems

Facts:

- Exponential lower bound for CP
  [Pudlák]
- Quasipolynomial simulation of treelike Res(CP*) in CP
  [Fleming et al.]
- Res(LP*)=Res(CP*)
  [H, Kojevnikov]

star* for unary coefficients

Open:

- Exponential lower bounds for Res(LP*)?
- Exponential lower bounds for treelike Res(LP)?

Tropical (min-plus) arithmetic

Min-plus arithmetic over $\mathbb{Q}_\infty$:

$$a \oplus b = \min(a, b), \qquad a \odot b = a + b$$

Tropical monomial:

$$m = x_1^{d_1} \odot x_2^{d_2} \odot \ldots \odot x_k^{d_k}$$

Tropical term:

$$t = c \odot m, \quad c \in \mathbb{Q}$$

The empty monomial (term), ∞ plays the role of zero:

$$\infty$$

Tropical polynomial:

$$p = t_1 \oplus t_2 \oplus \ldots \oplus t_m$$

Min-plus inequality:

$$p_1 \leqslant p_2$$

In the usual terms (for linear functions L_i):

$$\min(L_1, \ldots, L_m) \leqslant \min(L'_1, \ldots, L'_s)$$

or

$$L_1 \leqslant L \ \vee \ L_2 \leqslant L \ \vee \ \ldots \ \vee \ L_m \leqslant L$$

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Tropical proof systems

Input system of min-plus inequalities: $f_i \leqslant g_i$.

Min-Plus Nullstellensatz (MP-NS) — complete by [Grigoriev, Podolskii].

Multiply $f_i \leqslant g_i$ by polynomials and take the sum:

$$\bigoplus_i f_i \odot p_i \leqslant \bigoplus_i g_i \odot p_i$$

so that for every monomial (including ∞),
its coefficient on the left is $>$ than on the right.

Min-Plus Polynomial Calculus (MP-PC).

Do it step by step.

- Take the sum $\oplus$.
- Multiply by a term.
- Substitute $p \leqslant q$ into $h \leqslant p \oplus g$ or $q \oplus g \leqslant h$.
- Thus compose an MP-NS-like inequality.

Tropical resolution rule.

$$\frac{t \oplus f \leqslant 0 \qquad t' \oplus f \leqslant 0}{(t \odot t') \oplus f \leqslant 0}, \text{ where } t, t' \text{ are terms.} \qquad (\odot\mathrm{RES})$$

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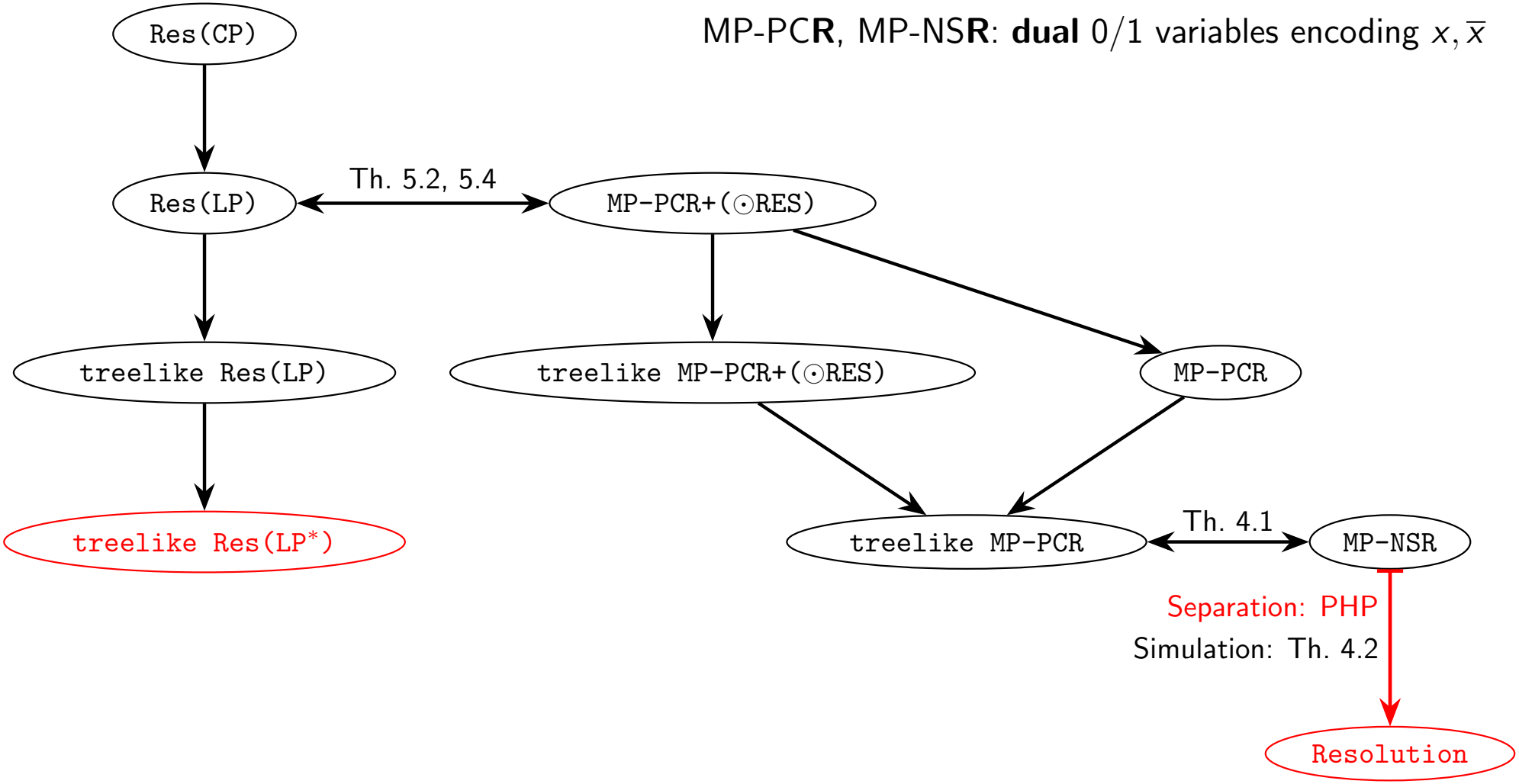
General picture

MP-PC**R**, MP-NS**R**: **dual** 0/1 variables encoding $x, \overline{x}$



Important: there is no equivalence to Res(LP) without (⊙RES), with some evidence.

PHP in `treelike Res(LP)` and `MP-NSR`

Treelike `Res(LP)` proof (motivated by the known treelike `CP` proof, but avoids rounding).

Lemma

$S_n := x_1 + \ldots + x_n$.
There is a `treelike Res(LP)` derivation of $S_n \leqslant 1$ from $x_i + x_j \leqslant 1$.

Proof.

- $S_{n-1} \leqslant 1$

  $\vdash S_{n-1} + x_n \leqslant 1 + x_n$.

- $x_i + x_n \leqslant 1$ (for $1 \leqslant i \leqslant n-1$)

  $\vdash S_{n-1} + (n-1)x_n \leqslant n-1$

  $\vdash S_{n-1} + x_n \leqslant 1 + (n-2)\overline{x}_n$.

- Add both inequalities to $x_n \leqslant 0 \lor (n-2)\overline{x}_n \leqslant 0$.

□

It does not use full (`+RES`), one premise is not a disjunction $\Rightarrow$ Switch to `MP-PCR`.

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PHP in `treelike Res(LP)` and `MP-NSR`

Switching to tropical Nullstellensatz (or `treelike MP-PCR`).

Lemma

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There is a `treelike MP-PCR` derivation of $S_n \leqslant 1$ from $x_i \odot x_j \leqslant 1$.

Proof.

- $S_{n-1} \leqslant 1$

  $\vdash S_{n-1} \odot x_n \leqslant 1 \odot x_n$.

- $x_i \odot x_n \leqslant 1$ (for $1 \leqslant i \leqslant n-1$)

  $\vdash S_{n-1} \odot x_n^{n-1} \leqslant n-1$

  $\vdash S_{n-1} \odot x_n \leqslant 1 \odot \overline{x}_n^{n-2}$.

- Take the tropical sum $S_{n-1} \odot x_n \leqslant (1 \odot x_n) \bigoplus (1 \odot \overline{x}_n^{n-2})$
  and substitute $(1 \odot x_n) \bigoplus (1 \odot \overline{x}_n^{n-2}) \leqslant 1$ □

Now it's a treelike `MP-PCR` refutation.
Some technicalities are needed for $x \oplus \overline{x}^{n-2} \leqslant 0$.

Daglike Resolution in MP-NSR

Translate $A = l_1 \vee \ldots \vee l_k$ into $[A] = 0 \odot l_1 \odot \ldots \odot l_k$.

Input clauses

$$1 \leqslant 0 \odot [C_i]. \tag{1}$$

At step $i = 1, 2, \ldots, s$, let $c_i = 1 - 1/(i+1)$.

Resolution

$$\frac{A \vee x \quad A \vee \neg x}{A},$$

becomes

$$c_i \odot x \odot [A] \bigoplus c_i \odot \overline{x} \odot [A] \leqslant c_i \odot [A]. \tag{2}$$

Weakening

$$\frac{A}{A \vee \ell},$$

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$$c_i \odot [A] \leqslant c_i \odot [A] \odot [\ell]. \tag{3}$$

Take the tropical sum of all these (1), (2), (3). They combine nicely!

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Encoding of Boolean logic

- Dual encoding: axioms

$$x \odot \overline{x} = 1, \qquad x \oplus \overline{x} = 0$$

- A Boolean clause:

$$\overline{x}_1 \oplus \overline{x}_2 \oplus \ldots \oplus \overline{x}_n \leqslant 0$$

- Or, like in CP:

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- Economic encoding:

$$x^2 \oplus x = 1$$

- Also $(0, \infty)$-encoding: turns MP-PC into Res(∞)

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Lower bounds

- Linear degree of MP-NSE refutations of the knapsack
$$\bigoplus_{1 \leqslant i \leqslant n} x_i = r < 0$$
- Exponential size bound:

  MP-NSR refutation size for the binomial $x^{\odot k} = c$ is greater than k
- Non-integer coefficients are needed in MP-NSE
- Tropical resolution ($\odot$RES) is not derivable in MP-PCR
- $0 \leqslant x$ is not derivable in MP-PCE (yet it is complete!)

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Open questions

1. Separate or simulate `Res(LP)` in `MP-PCR` without ($\odot$`RES`). Unary coefficients?
2. Quasipolynomially simulate `treelike Res(LP)` or `MP-NSR` in `CP`.
3. Simulate `Resolution` in `MP-NSE` (economic encoding).
4. Relations between `MP-NSR` and `CP` are unclear, both for unary and binary coefficients, and even for `treelike CP`.
5. Does adding the integer negation (as in `Res(CP)`) to `MP-NSR` allow it to simulate at least `treelike CP`?
6. Show exponential lower bounds for `MP-NSR` or `MP-NSE` (economic encoding) for CNFs.

(There are more open questions...)